BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE423: Computer Graphics

Assignment 03 | Total Marks: 20 | Summer 2025

Submission: August 20, 2025

Q1.	**a.**	Why is perspective projection considered more realistic for simulating human vision in 3D graphics, whereas parallel projection is often preferred in engineering and architectural drawings? Support your explanation with relevant examples. [2]
	b.	During the PUBG Mobile Campus Championship (PMCC) in BRAC University, the live-match analysis team is tracking the position of a player on the 3D battle map. At one moment, the player's location in the game world is **P(10, 20, -40)**, where X is the east–west direction, Y is the north–south direction, and Z is the elevation. To display the position on the overhead XY tactical screen, the team uses Cavalier Projection with an orientation angle of 30°. Calculate the projected point P′ on the xy plane. [4]
	c.	In a projection setup, the origin lies on the projection plane (PP), and the Center of Projection (COP) is 150 units away from the PP along the negative **y**-axis. The projection plane is aligned with the xz plane. Given a 3D point P (40, 75, -250), compute the projected coordinates **P′** on the projection plane. [4]

Q2.	**a.**	An architect is preparing a 3D visualization of a small wooden cabin. In the illustration, the front wall is drawn in its actual proportions, while the cabin's length is represented along lines inclined at 45° to the horizontal, with no reduction in length scale. Identify which type of parallel projection is applied here, and justify your answer by describing **two** defining properties of this

		projection method. What would be the effect on the visual appearance if the length axis were reduced to **half** its actual length? [3+1]
	b.	During the CSE425 quiz, Mr. Rahman is standing at the back of the classroom. In front of him is a transparent glass panel located at z = 120. Through the panel, he can see a whiteboard fixed at the front wall of the room, which extends from the point (200, 150, 360) to the point (600, 130, 360). A particular marker spot on the whiteboard is located exactly **40%** of the way from the first point toward the second point. Mr. Rahman's eye position is at (180, 300) in the x and y coordinates. The distance from his eye to the glass panel along the z-axis is **one-third** of the distance from the panel to the marker spot. Find the projected coordinate of the marker spot, P′, on the glass panel. [6]